

sync_all_markdown_exports.sh

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Companion user guide for `scripts/testing/sync_all_markdown_exports.sh` (§11.4.18 script documentation mandate).

Overview

Closes the §11.4.65 Universal Markdown export gap for the HelixVPN spec tree. For every `*.md` document under `docs/research/mvp/final/` (recursive) it generates two synchronized sibling artifacts:

- `<name>.html` — via pandoc (standalone, self-contained, embedded resources, document title taken from the source's first H1).
- `<name>.pdf` — via weasyprint, rendered from the generated HTML.

The script is idempotent: a sibling that is already newer than its source `.md` is skipped unless `--force` is given. Conversion runs sequentially (concurrency = 1) to respect the §12.6 60 % memory ceiling, with a timeout 60 per file per format.

Prerequisites

- pandoc on PATH (HTML generation). Verified: pandoc 3.9.x.
- weasyprint on PATH (PDF generation). Verified: WeasyPrint 66.x.
- bash, find, and a timeout/gtimeout binary (optional — the script degrades gracefully to no per-call timeout if absent).

Usage

```
# Whole spec tree (docs/research/mvp/final/, recursive)
bash scripts/testing/sync_all_markdown_exports.sh
```

```
# Limit to a sub-path (relative to repo root, or absolute)
bash scripts/testing/sync_all_markdown_exports.sh docs/research/
    mvp/final/v02-data-plane

# Force regeneration even when siblings are newer than the source
bash scripts/testing/sync_all_markdown_exports.sh --force
```

Final line of output:

```
generated=N skipped=M failed=K
```

Exit code 0 when $K == 0$, exit 1 when any file/format failed (the failed source paths are also printed to stderr).

Edge cases

- **Per-file failure does not abort the run.** A pandoc or weasyprint failure on one document is logged to stderr, the run continues, and the process exits non-zero at the end so the failure is not silently lost (§11.4.1 — no FAIL-bluff, no silent skip).
- **Concurrent spec generation.** The file list is snapshotted by a single find at start. If another process adds .md files mid-run, those late arrivals are not in the snapshot — simply re-run the script (idempotent) to generate siblings for them; already-current siblings are skipped.
- **mtime skip.** A sibling is skipped only when it is strictly newer than its .md source. Editing the source makes both siblings stale and they regenerate on the next run.
- **Title extraction.** The first ATX # H1 line is used as the pandoc document title; if none is present the basename is used.
- **Mermaid code fences (§11.4.168 known limitation).** Plain pandoc does NOT render ````mermaid blocks into diagrams — they are emitted as `<pre class="mermaid"> source text in the HTML`, and therefore appear as raw diagram source (not a rendered picture) in the PDF. Rendering Mermaid requires a pandoc filter (e.g. mermaid-filter / mermaid-cli), which is out of scope for this wrapper. Until such a filter is wired in, the HTML/PDF carry the prose faithfully but show Mermaid diagrams as code, not images. Do not claim visual diagram fidelity for these blocks.

Internal behaviour

1. Resolve repo root from the script location, set scope root to docs/research/mvp/final.
2. Parse args: optional path-prefix (limit scope) and --force.
3. Verify pandoc + weasyprint are present; abort with a clear error otherwise.

4. find all *.md under the walk root into a newline-delimited temp file (spec doc paths contain no newlines), iterate with `while IFS= read -r`.
5. For each source: compute .html/.pdf sibling paths, apply the mtime skip unless `--force`, then run pandoc (HTML) and weasyprint (PDF) under `timeout 60`.
6. Tally generated / skipped / failed, print the summary, exit 0/1.

The script is written POSIX-ish and parses cleanly under BOTH `sh -n` and `bash -n` (§11.4.67) — no bash-only constructs (process substitution, NUL handling, `[ [ ]]`, arrays) are used.

Related scripts

- This is a project-level instantiation of the §11.4.65 export discipline for the spec tree. A full-repo equivalent (`docs/**`, owned-submodule READMEs, etc.) is the broader §11.4.65 mandate; this wrapper covers the `docs/research/mvp/final/ MVP` specification corpus specifically.

Last verified date

2026-06-26 — full run over `docs/research/mvp/final/` produced `generated=75 skipped=14 failed=0`; an idempotent follow-up run closed the gap from concurrently-added spec files (`md = html = pdf` counts equal, zero missing siblings). One generated PDF was confirmed to carry real rendered prose (20 pages, title + revision header + body) via `pypdf`; the matching HTML carried rendered `<h1>/<h2>` heading tree + ToC. Mermaid blocks confirmed to remain as `<pre class="mermaid">` source (see Edge cases).